

FRONT_BOT (Frontline Reporting & OSINT Node Tracker Bot)

Developed by Zsolt Lazar – OSINT Homelab

FRONT_BOT

Frontline Reporting & OSINT Node Tracker Bot

Version 2.0

Military conflict	Russian invasion of Ukraine
Opposing parties	Russia – Ukraine
Conflict start date	24.Feb.22
Conflict day	1393
Conflict week	199
Reporting period	2025-12-08 → 2025-12-15
GitHub system info	Linux-6.11.0-1018-azure-x86_64-with-glibc2.39 CPU: x86_64 RAM: 8.33 GB
Reporting FRONT_BOT version	2.0
Report writing duration	4 minutes 18 seconds
Project developer	Zsolt Lazar
Developer's website	https://bio.site/zsoltlazar
Project website	https://frontbot.my.canva.site/front-bot

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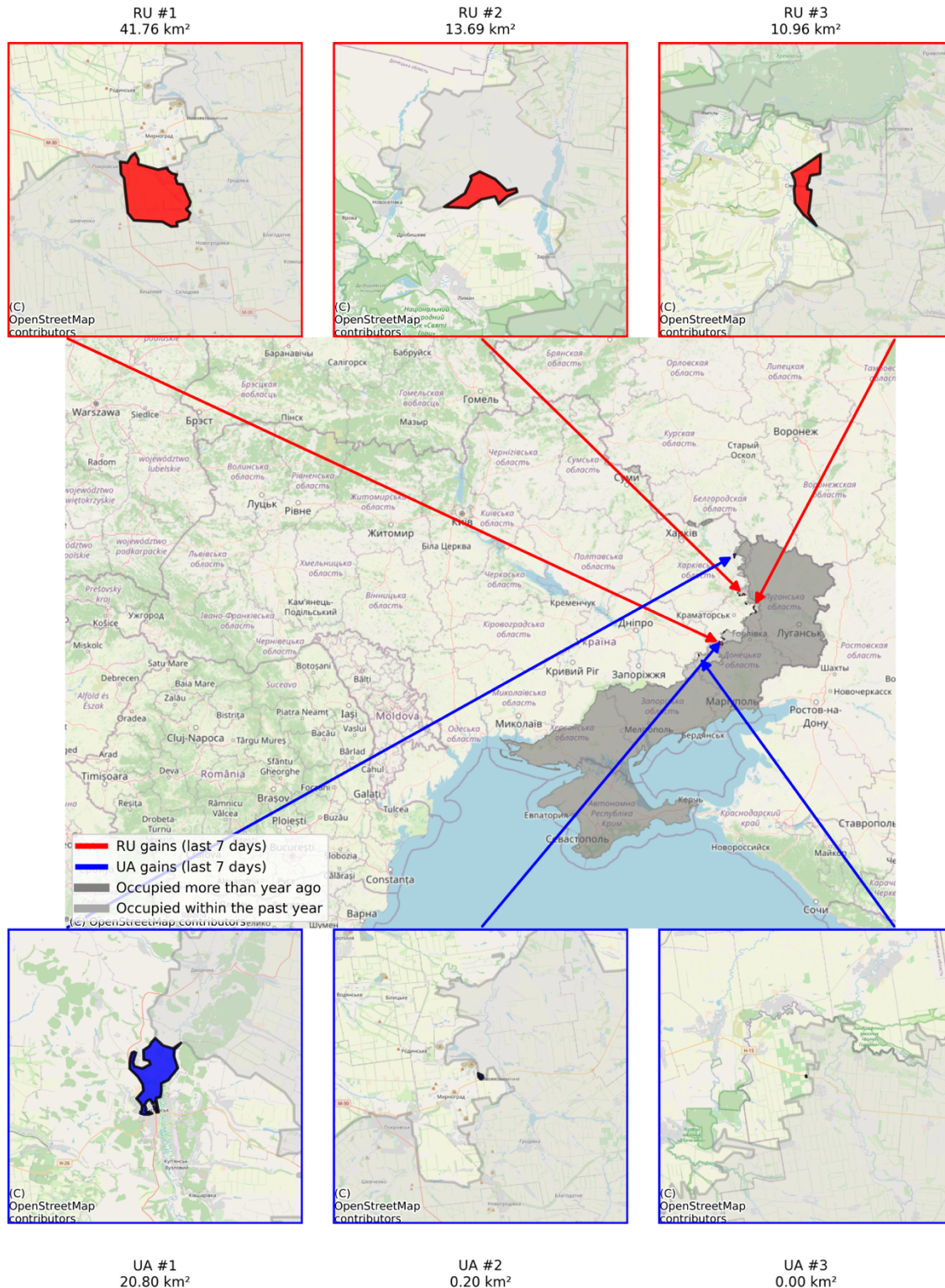
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Weekly Territorial Map (DeepStateMap)

Last 7 Days | RU Gains 103.77 km² | UA Gains 21.00 km²
Top 3 territorial changes shown on side maps



FRONT_BOT is a research prototype for automated military conflict reporting.
Powered entirely by publicly available OSINT datasets and Python-based data analysis pipelines.
⚠ All data is interpreted algorithmically — inaccuracies may occur. Use at your own discretion.

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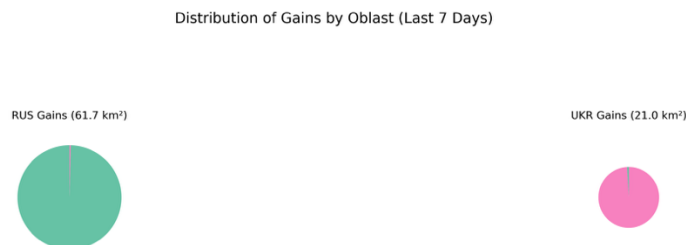
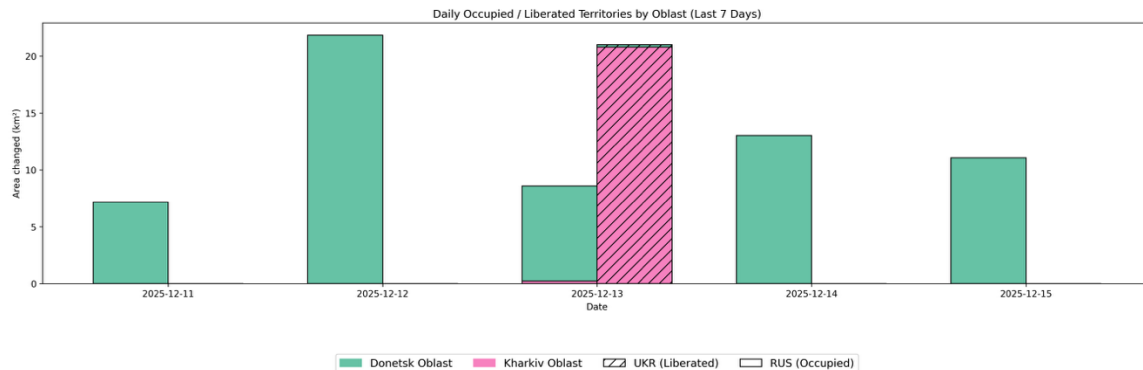
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Executive Summary

During the reporting period, Russian forces sustained a high operational tempo across multiple axes, achieving net territorial gains at both the operational and tactical levels, particularly in Donetsk Oblast and within the urban area of Pokrovsk. Although the weekly rate of territorial expansion declined compared to the previous reporting cycle, Russian forces continued to consolidate control in key sectors while applying sustained pressure on Ukrainian defensive positions. Ukrainian forces achieved localized counter-successes, most notably within Kupiansk, where a significant net territorial recovery was recorded. Equipment loss data indicates continued high-intensity combat, with both sides experiencing losses well above long-term weekly averages, reflecting ongoing attritional engagements along active fronts.

Operational Overview

According to DeepStateMap data covering the period from 08 to 15 December 2025, Russian forces gained a total of 103.77 square kilometers over seven days. The three largest territorial gains accounted for 66.40 square kilometers, representing approximately 64 percent of all Russian advances during the period. Ukrainian forces, by comparison, achieved total gains of 21.00 square kilometers, with all gains concentrated within the three largest advances. The overall balance of territorial change during this interval therefore remained decisively in Russia's favor.

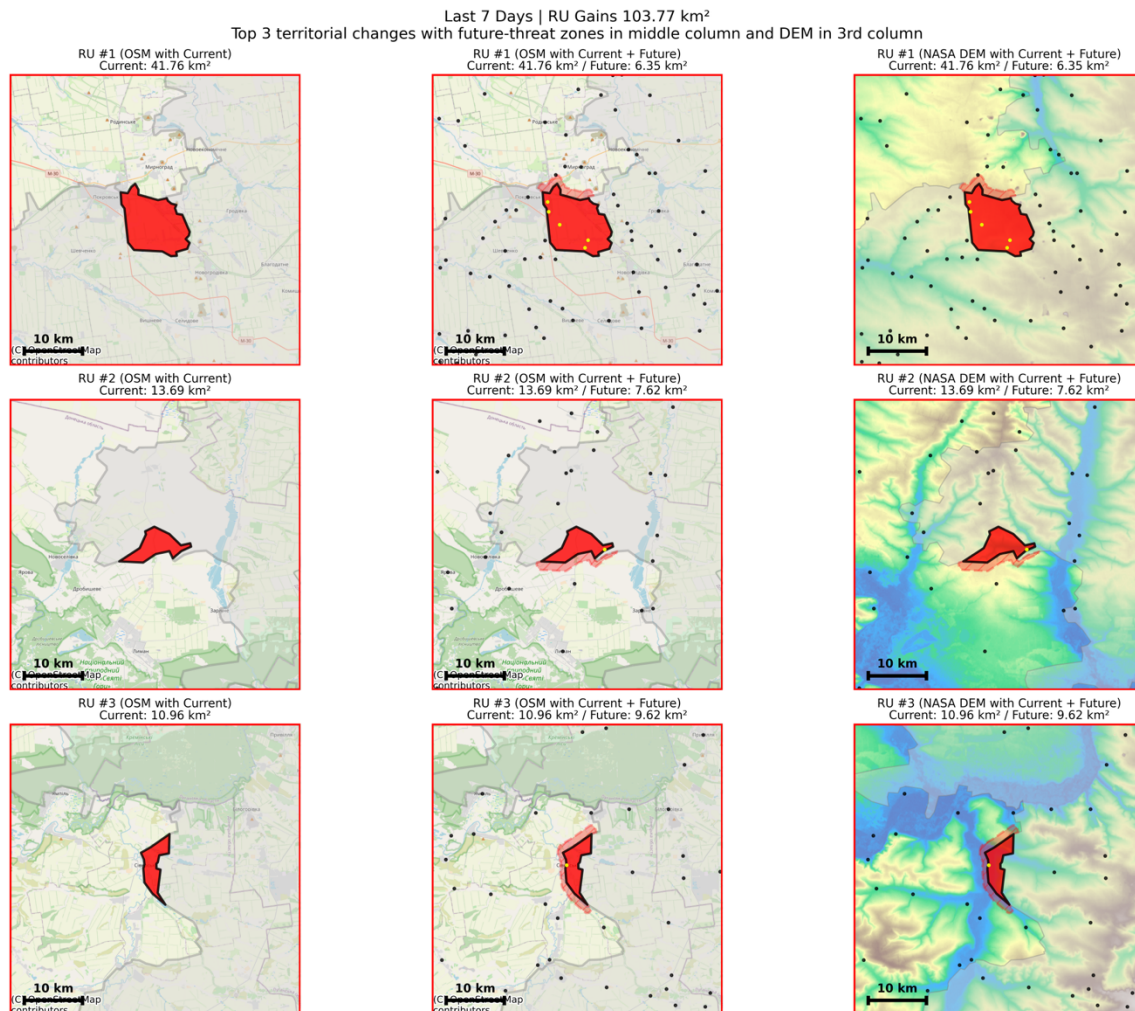


The largest single Russian advance during the week covered approximately 41.76 square kilometers, accounting for roughly 40 percent of total Russian gains. This advance affected the settlements of Rih, Sukhyi Yar, Novopavlivka, Hnativka, and Lysivka. Should the current rate and direction of advance be sustained, the settlement of Chunyshyne is assessed as likely to enter the contested zone in the near term. The second-largest Russian gain amounted to approximately 13.69 square kilometers, affecting the settlement of Kolodiaz, with no immediate spillover anticipated. The third-largest advance covered 10.96 square kilometers and

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affected the town of Siversk, where the frontline remains relatively stable following the advance.



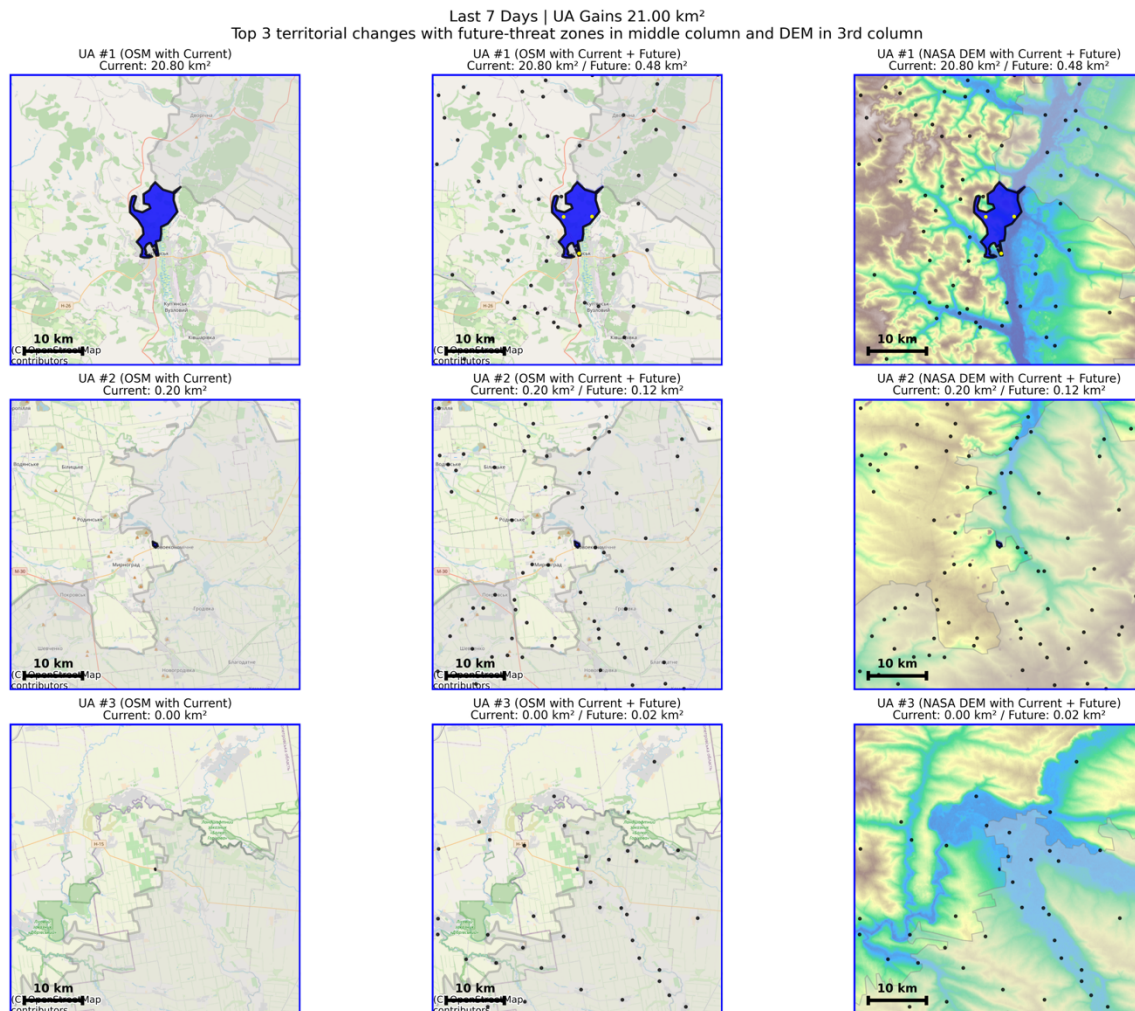
Ukrainian territorial gains between 08 and 15 December totaled 21.00 square kilometers. The most significant Ukrainian advance, measuring approximately 20.80 square kilometers and accounting for 99 percent of total Ukrainian gains, affected the settlements of Holubivka, Radkivka, and Kupiansk. Two additional minor Ukrainian advances were recorded, measuring 0.20 square kilometers and 0.00 square kilometers respectively, neither of which affected any settlements nor indicated immediate further expansion.

A broader assessment covering 09 to 15 December 2025 indicates that Russian forces expanded control by 237.1 square kilometers, representing a 19.7 percent decrease in weekly gains compared to the previous week. Ukrainian forces gained 1.4 square kilometers over the same interval, reflecting an 89.7 percent reduction compared to the preceding reporting period. The geographic focus of Russian expansion remained overwhelmingly concentrated in Donetsk Oblast, where a series of settlements—including Vyiimka, Siversk, Tolstoi, Pankivka, Maiak, Rih, Sukhyi Yar, Novopavlivka, Hnativka, Lysivka, Mykolaivka, and Kolodiaz—were assessed as the most heavily occupied or newly occupied during the period. Newly occupied locations consisted

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entirely of villages and towns within Donetsk Oblast, underscoring the continued prioritization of this operational direction.



Based on a median weekly Russian advance of approximately 0.67 kilometers and a dynamic buffer of 672 meters, Novoekonomichne, Volodymyrivka, and Fedorivka in Donetsk Oblast are assessed as the settlements most likely to be affected by further Russian advances. Conversely, based on negligible Ukrainian weekly retreat and a dynamic buffer of 500 meters, Kupiansk in Kharkiv Oblast remains the settlement most likely to experience further Ukrainian counter-advances.

Equipment Losses and Attrition

WarSpotting data for the period from 09 to 15 December indicates that Russian forces lost 13 pieces of equipment, bringing cumulative confirmed losses to 22,018. This represents a 61.8 percent reduction compared to the previous week and remains significantly below the long-term weekly average of 110.1 losses. The most notable relative increase occurred in anti-aircraft systems, which rose to two losses, while transport losses declined sharply to two units, an 80 percent decrease week-on-week. Equipment categories showing the highest relative increase

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compared to their yearly averages included “Other” equipment, command and communications assets, and anti-aircraft systems. Categories experiencing the largest declines included aircraft, ambulances and medical vehicles, and anti-tank systems, all of which recorded zero losses during the period.

Geolocation data was available for 46.2 percent of reported Russian equipment losses, down from 67.6 percent in the previous week. The 76th Guards Air Assault Division recorded the highest number of identifiable equipment losses among Russian units. Pokrovsk, within Pokrovsk Raion, emerged as the most frequently reported loss location, indicating sustained or intensified combat activity in and around the urban area.

Oryx-verified data covering the same period shows substantially higher reported losses for both sides, reflecting differences in methodology and confirmation criteria. Russian forces lost 118 units, raising cumulative losses to 23,669, while Ukrainian forces lost 166 units, bringing their cumulative total to 10,794. These figures represent week-on-week increases of 0.5 percent for Russia and 1.6 percent for Ukraine. The Russian-to-Ukrainian loss ratio declined slightly from 2.22 to 2.19. Both sides recorded weekly losses well above their respective long-term averages, indicating a period of intensified combat and elevated attrition.

Within the Oryx dataset, Russian losses were particularly pronounced in vehicles, armor, infantry, IFVs, and tanks, all of which exceeded historical weekly averages by a substantial margin. Ukrainian losses were dominated by infantry, infantry mobility vehicles, APCs, and artillery, again significantly exceeding baseline levels. These patterns suggest sustained combined-arms engagements with heavy reliance on infantry-supported maneuver and logistical movement on both sides.

As of 15 December 2025, DeepStateMap assesses that Russian forces occupy approximately 96,187.83 square kilometers of Ukrainian territory, corresponding to 16.00 percent of Ukraine’s total land area. Over the preceding seven days, occupied territory increased by 82.77 square kilometers, representing a marginal growth of 0.01 percent. Luhansk Oblast remains almost entirely occupied at 99.52 percent, while Crimea remains at 98.96 percent, with no changes recorded in either region during the week. Donetsk Oblast saw the most significant change, with occupied territory increasing by 103.34 square kilometers, raising occupation to 77.13 percent of the oblast.

At the current aggregate weekly pace, full occupation of Ukraine would theoretically require approximately 6,100 weeks, or roughly 117 years, underscoring the strategic limitations of incremental territorial expansion absent a major operational breakthrough. Among partially occupied oblasts, Donetsk Oblast remains the most critical, with current trends suggesting potential full occupation within approximately 59 weeks if the current rate of advance is sustained. Zaporizhia Oblast remained static at 74.6 percent occupied, while Kharkiv Oblast saw a net reduction in occupied territory of 20.6 square kilometers, indicating stalled or reversed Russian progress in that sector.

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Settlement level

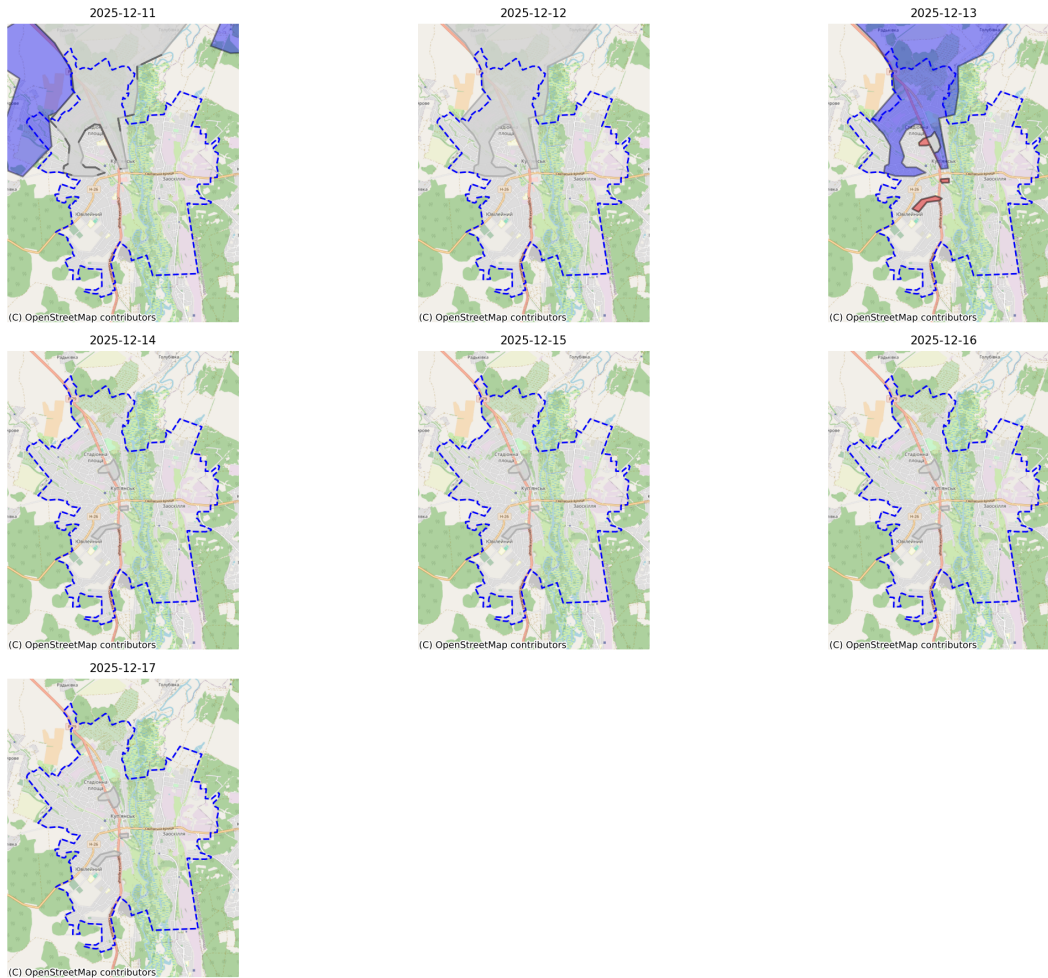
Urban-level analysis highlights diverging trends in two key contested cities. Within Pokrovsk, between 11 and 17 December, Russian forces increased occupied territory by 2.24 square kilometers, bringing total occupation to 19.39 square kilometers, or 62.2 percent of the settlement's total area. The largest single-day gain occurred on 14 December, while no Ukrainian territorial recovery was recorded during the period, resulting in a net shift in favor of Russian forces.



In contrast, within Kupiansk, Russian forces gained only 0.23 square kilometers, while Ukrainian forces recaptured 5.22 square kilometers over the same period. This resulted in a net territorial change of minus 4.99 square kilometers for Russian forces and reduced Russian occupation of Kupiansk to 22.9 percent of the settlement area, marking a notable Ukrainian tactical success.

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Conclusion

Overall, the reporting period reflects continued Russian territorial expansion concentrated in Donetsk Oblast and around Pokrovsk, balanced by localized Ukrainian counter-offensives in Kharkiv Oblast, particularly within Kupiansk. High equipment losses on both sides indicate sustained combat intensity, while territorial trends suggest a gradual but uneven consolidation of Russian control rather than rapid operational breakthroughs.

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Equipment Losses Weekly Comparison (WarSpotting)

Operational Overview

During the reporting period, confirmed equipment losses and territorial developments indicate a continued high-intensity operational environment characterized by attritional engagements, localized maneuver, and uneven territorial dynamics. While Russian equipment losses recorded by WarSpotting declined sharply compared to the previous week, Oryx-verified data suggests substantially higher overall attrition on both sides, highlighting methodological differences and ongoing combat intensity. Territorial changes remain concentrated in eastern Ukraine, with Donetsk Oblast continuing to absorb the majority of Russian expansion, while Ukrainian forces achieved a measurable territorial reversal in Kharkiv Oblast.

Module 1 – Equipment Losses (WarSpotting)

Between 09 and 15 December 2025, WarSpotting recorded **13 Russian equipment losses**, representing a **61.8% decrease** compared to the previous week (–21 units). Losses during this period were well below the long-term weekly average, suggesting either reduced exposure of high-value systems, improved force protection, or reduced reporting density in active sectors.

The loss profile shifted toward air-defense and command-related assets, with **anti-aircraft systems and command-and-communication nodes** registering increases from zero in the prior week. In contrast, losses of **tanks, engineering equipment, drones, self-propelled artillery, and transport vehicles** declined sharply, with several categories recording zero confirmed losses. Transport losses, previously dominant, dropped by 80%, indicating reduced interdiction of logistical movements or improved dispersal.

Geolocation coverage declined to **46.2%**, down from **67.6%** the previous week, reducing spatial confidence in loss attribution. Identifiable unit-level losses were minimal, with only a single confirmed loss linked to the **76th Guards Air Assault Division**. Pokrovsk and surrounding areas remained the most frequently reported loss locations, reinforcing assessments of sustained combat activity in the Donetsk operational sector.

Table 1 – WarSpotting Weekly Equipment Losses (Russia)

Equipment Type	Losses (This Week)	Change vs Previous Week	% Change
Anti-aircraft systems	2	+2	+0.0%
Command posts / Communication	1	+1	+0.0%
Drones	0	-2	-100.0%
Engineering	0	-3	-100.0%
Infantry fighting vehicles	4	-3	-42.9%
Other	3	-1	-25.0%
Rocket & missile artillery	1	0	+0.0%
Self-propelled artillery	0	-1	-100.0%
Tanks	0	-6	-100.0%
Transport	2	-8	-80.0%
Total	13	-21	-61.8%

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Geolocation and Attribution Summary

Metric	Value
Geolocated losses (current week)	6 / 13 (46.2%)
Geolocated losses (previous week)	23 / 34 (67.6%)
Most affected unit	76th Guards Air Assault Division (1 loss)
Top locations	Pokrovsk (2), Donetsk Oblast (1), Kramatorsk Raion (1)

Module 2 – Equipment Losses (Oryx)

Oryx data for the same period indicates **significantly higher attrition levels** for both belligerents. Russian cumulative losses increased by **118 units** to **23,669** (+0.5%), while Ukrainian cumulative losses rose by **166 units** to **10,794** (+1.6%). The **RU/UA loss ratio** decreased slightly from **2.22** to **2.19**, indicating marginally higher relative Ukrainian losses during the period. Russian losses were broadly distributed across nearly all equipment categories, with particularly high counts in **vehicles, armor, infantry, IFVs, and tanks**, reflecting sustained combined-arms engagement. Ukrainian losses were dominated by **infantry, infantry mobility vehicles, APCs, and artillery**, indicating intense defensive and counter-mobility operations, as well as exposure of personnel-heavy formations.

Table 2 – Oryx Weekly Equipment Losses (Russia)

Category	Weekly Losses
Tanks	18
AFV	14
IFV	26
APC	5
IMV	0
Engineering	6
Communications	0
Vehicles	34
Aircraft	2
Infantry	31
Logistics	6
Armor	32
Antiair	2
Artillery	1

Table 3 – Oryx Weekly Equipment Losses (Ukraine)

Category	Weekly Losses
Tanks	8
AFV	4
IFV	10

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APC	22
IMV	50
Engineering	0
Communications	4
Vehicles	2
Aircraft	1
Infantry	82
Logistics	4
Armor	12
Antiair	0
Artillery	18

Cumulative Totals

Side	Cumulative Losses	Weekly Change	% Change
Russia	23,669	+118	+0.5%
Ukraine	10,794	+166	+1.6%
RU/UA Ratio	2.19	↓ from 2.22	—

Module 3 – Occupation Change Analysis (DeepStateMap)

Between 08 and 15 December 2025, total Russian-occupied territory in Ukraine increased by **82.77 km²**, corresponding to a **0.01 percentage-point** rise in overall occupation. Territorial change remained heavily skewed toward **Donetsk Oblast**, which alone accounted for a net increase of **103.34 km²**. This expansion was partially offset by a **20.57 km² reduction** in Russian-held territory in **Kharkiv Oblast**, indicating localized Ukrainian counter-pressure. No meaningful territorial changes were recorded in **Luhansk Oblast** or the **Autonomous Republic of Crimea**, both of which remain near-total occupation saturation.

Table 4 – Occupation Change by Oblast

Oblast	Change (km ²)	Percent Change
Donetsk	+103.34	+0.39%
Kharkiv	-20.57	-0.06%
Luhansk	0.00	0.00%
Crimea	0.00	0.00%
Total (Ukraine)	+82.77	+0.01 pp

Oblasts with Significant Occupation (>10%)

Oblast	% Occupied
Luhansk	99.52%
Autonomous Republic of Crimea	98.96%
Donetsk	77.13%
Zaporizhia	74.64%

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Module 10 – Settlement level (DeepStateMap)

Pokrovsk

Table 1 – Settlement Overview (Pokrovsk)

Parameter	Value
Settlement	Покровськ
Data Source	DeepStateMap
Reporting Period	2025-12-11 to 2025-12-17
Total Settlement Area (km ²)	31.17
Occupied Area (km ²)	19.39
Occupied Share (%)	62.2
Frontline First Entry Date	2025-09-29

Table 2 – Weekly Territorial Change Summary (Inside Pokrovsk)

Metric	Russian Forces (RU)	Ukrainian Forces (UA)
Total Area Gained (km ²)	2.24	0.00
Largest Single-Day Gain (km ²)	1.19 (2025-12-14)	0.00
Smallest Single-Day Gain (km ²)	0.00 (2025-12-12)	0.00
Net Change (km ²)	+2.24	0.00

Table 3 – Daily Territorial Changes Inside Pokrovsk

Date	RU Gain (km ²)	UA Gain (km ²)	Net Daily Change (km ²)
2025-12-11	1.049	0.000	+1.049
2025-12-12	0.000	0.000	0.000
2025-12-13	0.000	0.000	0.000
2025-12-14	1.187	0.000	+1.187
2025-12-15	0.000	0.000	0.000
2025-12-16	0.000	0.000	0.000
2025-12-17	0.000	0.000	0.000

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Module 10 – Settlement level (DeepStateMap)

Kupiansk

Table 1 – Settlement Overview (Kupiansk)

Parameter	Value
Settlement	Kupiansk
Data Source	DeepStateMap
Reporting Period	2025-12-11 to 2025-12-17
Total Settlement Area (km ²)	24.47
Occupied Area (km ²)	5.61
Occupied Share (%)	22.9
Frontline First Entry Date	2025-07-26

Table 2 – Weekly Territorial Change Summary (Inside Kupiansk)

Metric	Russian Forces (RU)	Ukrainian Forces (UA)
Total Area Gained (km ²)	0.23	5.22
Largest Single-Day Gain (km ²)	0.23 (2025-12-13)	4.73 (2025-12-13)
Smallest Single-Day Gain (km ²)	0.00 (2025-12-12)	0.00 (2025-12-12)
Net Change (km ²)	−4.99	+4.99

Table 3 – Daily Territorial Changes Inside Kupiansk

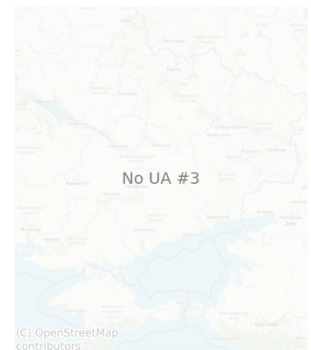
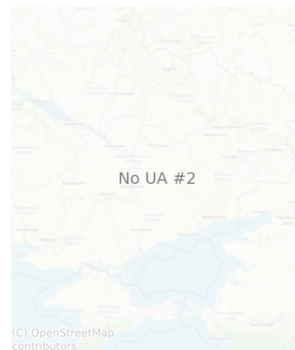
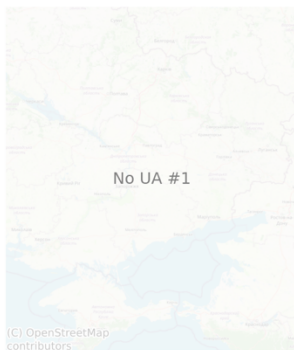
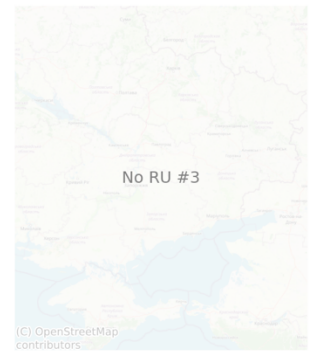
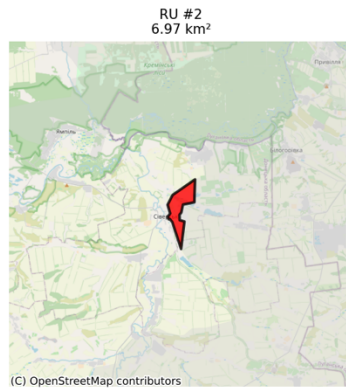
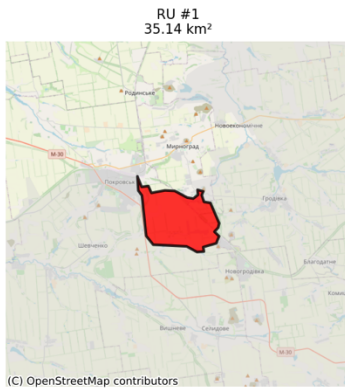
Date	RU Gain (km ²)	UA Gain (km ²)	Net Daily Change (km ²)
2025-12-11	0.000	0.491	−0.491
2025-12-12	0.000	0.000	0.000
2025-12-13	0.230	4.733	−4.503
2025-12-14	0.000	0.000	0.000
2025-12-15	0.000	0.000	0.000
2025-12-16	0.000	0.000	0.000
2025-12-17	0.000	0.000	0.000

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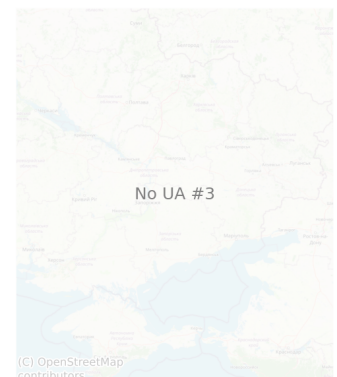
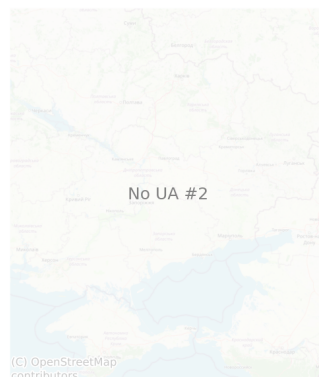
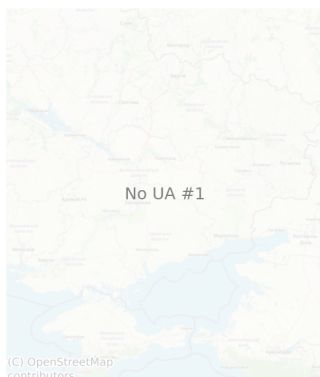
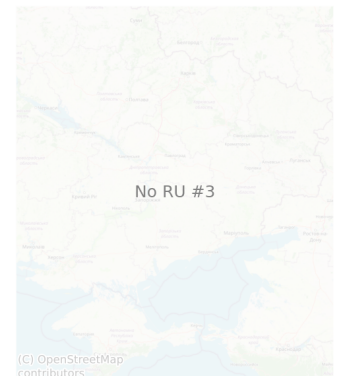
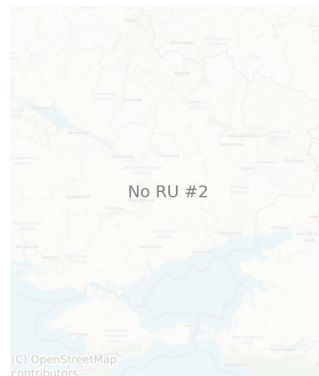
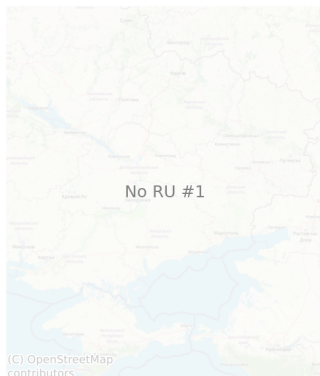
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Module 6: Frontline Map Report

2025-12-09 | Daily Total: RU: 42.11 km² | UA: 0.00 km² (Top 3 gains shown)



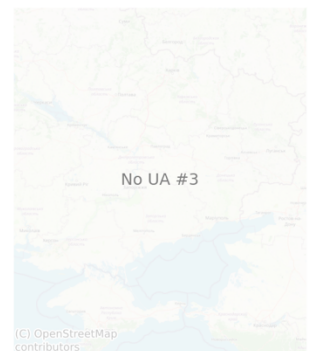
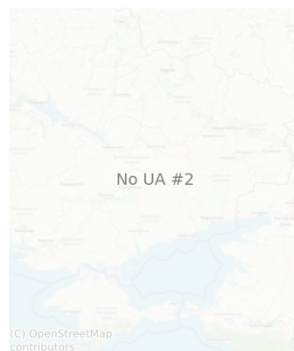
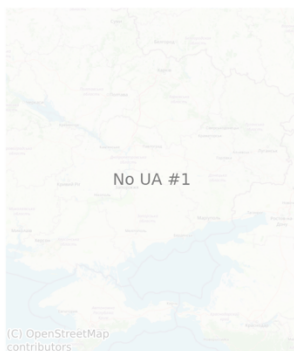
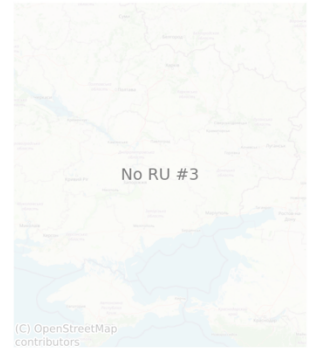
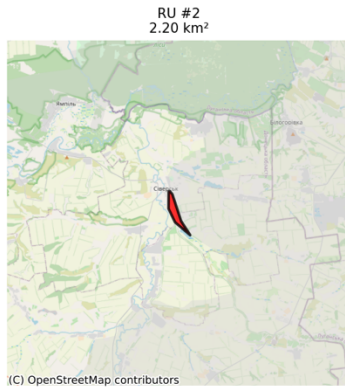
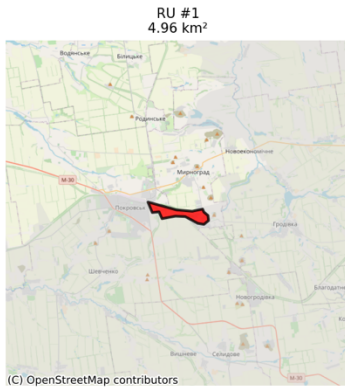
2025-12-10 | Daily Total: RU: 0.00 km² | UA: 0.00 km² (Top 3 gains shown)



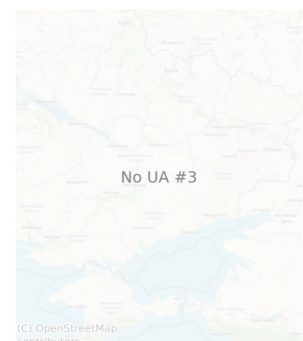
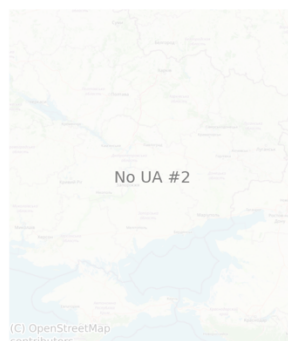
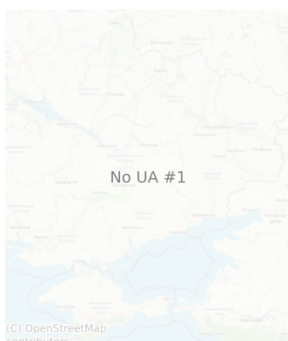
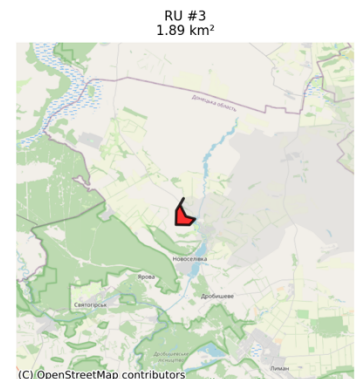
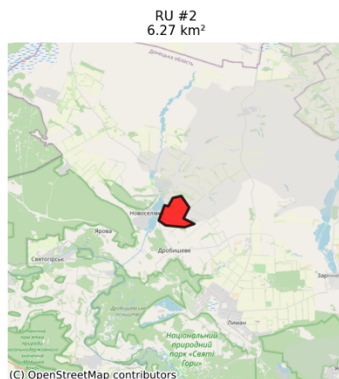
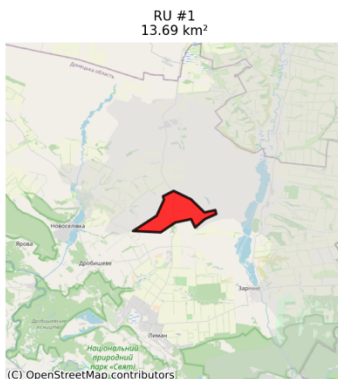
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2025-12-11 | Daily Total: RU: 7.16 km² | UA: 0.00 km² (Top 3 gains shown)



2025-12-12 | Daily Total: RU: 21.84 km² | UA: 0.00 km² (Top 3 gains shown)



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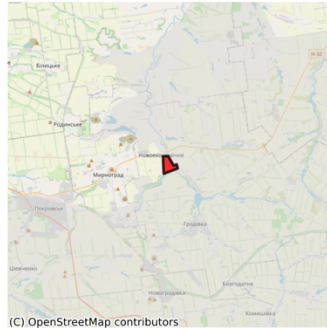
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2025-12-13 | Daily Total: RU: 8.58 km² | UA: 21.00 km² (Top 3 gains shown)

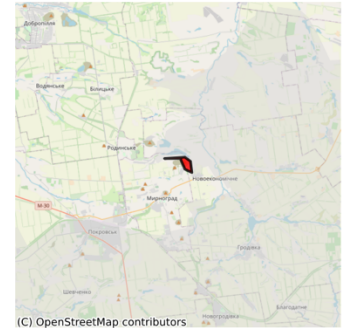
RU #1
3.51 km²



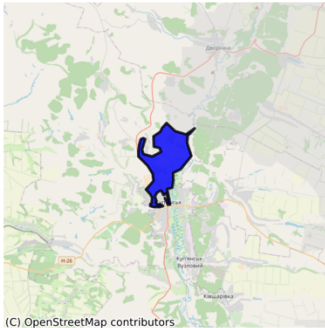
RU #2
1.61 km²



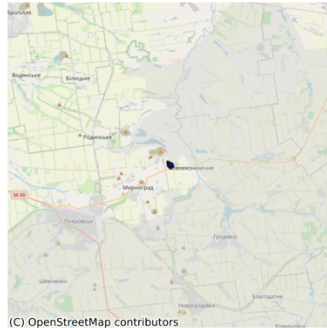
RU #3
1.13 km²



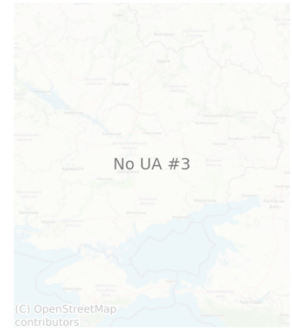
UA #1
20.80 km²



UA #2
0.20 km²

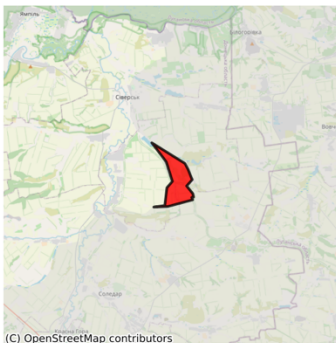


No UA #3

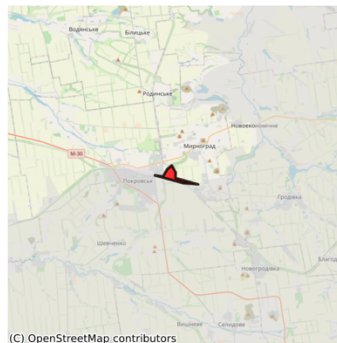


2025-12-14 | Daily Total: RU: 13.03 km² | UA: 0.00 km² (Top 3 gains shown)

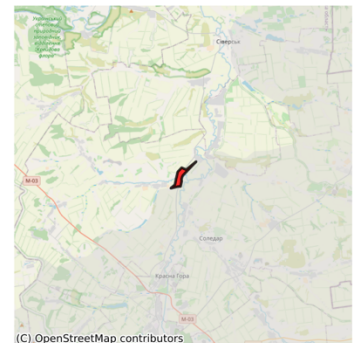
RU #1
10.13 km²



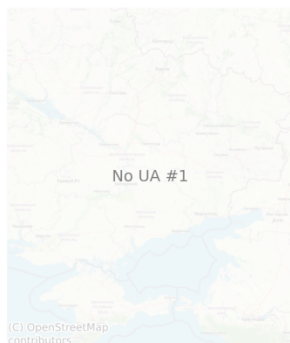
RU #2
1.65 km²



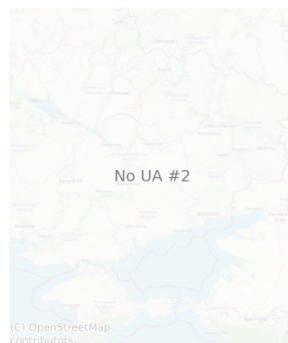
RU #3
1.25 km²



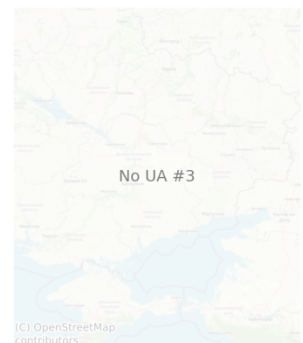
No UA #1



No UA #2



No UA #3

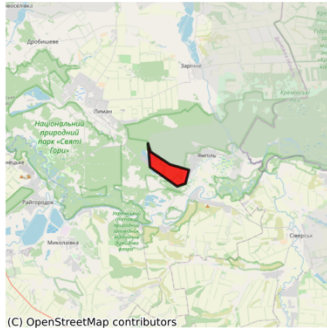


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2025-12-15 | Daily Total: RU: 11.06 km² | UA: 0.00 km² (Top 3 gains shown)

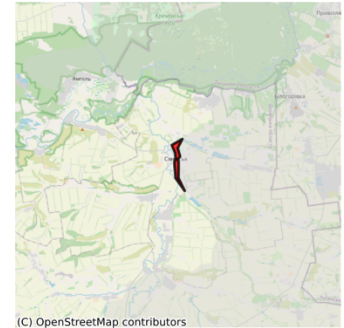
RU #1
6.59 km²



RU #2
2.67 km²



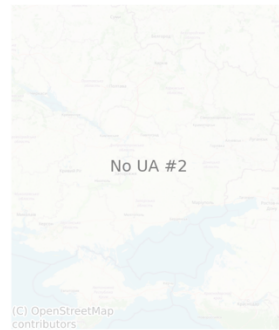
RU #3
1.80 km²



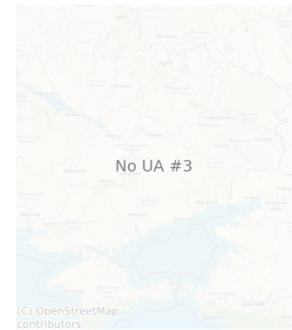
UA #1
0.00 km²



No UA #2



No UA #3



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METHODOLOGY

Module 1: Equipment Losses Weekly Comparison (WarSpotting) relies on open-source intelligence from the WarSpotting database, which aggregates visually confirmed Russian equipment losses reported during the conflict in Ukraine. The module uses an automated pipeline to download the most recent dataset from a controlled Kaggle repository if it is not already present locally. After ingestion, the data is cleaned by parsing the date field, removing invalid entries, and retaining key attributes such as equipment type, unit affiliation, nearest location, and geocoordinates where available. The analytical approach is based on a comparative weekly framework: the last seven-day period ending on the most recent entry is contrasted with the preceding week. For both timeframes, the module calculates total losses, the distribution of equipment types, the proportion of geolocated cases, and the most frequently affected units and locations. It then determines absolute and percentage changes in losses across weeks, identifying the largest increases or decreases by equipment category. Outputs are generated in two forms: a structured statistical summary presenting detailed counts and trends, and a natural-language narrative that highlights key findings in descriptive form.

Module 2: Equipment Losses Weekly Comparison (Oryx) performs a similar comparative analysis of equipment losses but uses the Oryx dataset, which is based on visually verified equipment losses for both Russian and Ukrainian forces. The data is sourced from an automated Kaggle repository and is structured as daily cumulative counts across multiple equipment categories for each side. After loading, dates are standardized and all numerical fields are converted to consistent numeric formats to ensure accuracy in time-series calculations. The analysis computes weekly increments by subtracting cumulative totals from the previous week, producing absolute and percentage changes for Russia and Ukraine separately. It also calculates the ratio of Russian to Ukrainian losses and compares these to the prior week to assess relative attrition trends. Beyond weekly differences, the module incorporates a retrospective component, analyzing up to one year of historical data to contextualize the current week's figures against long-term averages and identifying the most significant weekly spikes. The analysis extends to category-level dynamics, evaluating which equipment types recorded the highest increases or decreases relative to their historical averages, thereby identifying emerging tactical or operational trends. The results are formatted as two outputs: a structured statistical summary and a natural-language narrative enriched with retrospective context and category-level interpretation.

Module 3: Occupation Change Analysis (DeepStateMap) evaluates territorial control dynamics by comparing geospatial frontline data from DeepStateMap across a one-week interval. The module combines daily polygons of occupied areas with official oblast-level administrative boundaries, both projected into an equal-area coordinate system to enable precise areal calculations. Occupied and total oblast areas are computed through geometric intersections, producing the absolute extent of occupied territory in square kilometers as well as the percentage of each oblast under control. The analysis first establishes national-level occupation figures for the latest date and contrasts them with values from seven days earlier, calculating absolute and percentage changes to capture weekly territorial shifts. It then disaggregates results at the oblast level, ranking regions by occupation percentage and highlighting the three with the largest increases and decreases in occupied area. A significance filter is applied to identify oblasts with more than ten percent of their territory under occupation, providing a concise set of regions for targeted monitoring. The narrative consists of a short executive-style summary describing national-level occupation and change, and a more detailed report listing oblast-level increases, decreases, and significant cases.

Module 4: Settlements (DeepStateMap) extends the frontline analysis by linking geospatial control changes to populated places, providing a settlement-level perspective on the war's dynamics. The module integrates daily occupied-area polygons from DeepStateMap with point data of Ukrainian settlements (cities, towns, villages) and oblast boundaries derived from official administrative datasets. Geometries are reprojected into a metric coordinate system to ensure accurate distance and area measurements. Weekly gains and losses are computed by differencing occupied-area unions from the most recent seven-day period with those from the preceding week, yielding net Russian gains and Ukrainian recaptures in square kilometers. These figures are contextualized against the previous two weeks to calculate percentage changes in control dynamics. Newly occupied and newly liberated settlements are identified through geometric intersections of the frontline change polygons with settlement points, while oblast-level aggregation highlights the regions experiencing the most pronounced shifts in control. The analysis further estimates the average advance and retreat distances of frontline segments and applies dynamic buffers to detect settlements most at risk of being directly affected in the near term. The outputs consist of a narrative summary describing weekly territorial dynamics, settlement gains and losses, and oblast-level hotspots, supplemented by a forward-looking report listing threatened settlements based on proximity to recent frontline movement.

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Module 5: Occupation Projections (DeepStateMap) extends the static occupation analysis by introducing forward-looking estimates of territorial control. The module begins with daily DeepStateMap frontline polygons reprojected into an equal-area system and intersects them with oblast-level administrative boundaries. Occupied area and percentage are calculated for both the most recent and the preceding week, producing absolute and relative changes. These figures form the basis of simple linear projections: if an oblast's occupied area has increased, the weekly increment is extrapolated to estimate the time required for full occupation, expressed in both weeks and years. Conversely, where no gains are observed, occupation is assessed as stalled and no projection is generated. The results are organized hierarchically, beginning with national-level occupation dynamics and continuing with oblast-level projections for regions with partial occupation—defined as between two and ninety-five percent. This filtering ensures that analytical attention is focused on contested territories rather than those almost fully occupied or almost entirely liberated. Outputs consist of structured CSV tables with current, past, and projected values, accompanied by a narrative report. The narrative provides an OSINT-style briefing summarizing overall occupation trends, highlighting oblasts with significant changes, and discussing likely trajectories under current conditions. While the projections offer intuitive insights into possible future scenarios, they remain conditional, assuming the continuation of short-term trends. As with all extrapolations, the results are sensitive to fluctuations in operational tempo, sudden offensives, or withdrawals, and should therefore be treated as indicative rather than predictive.

Module 6: Maps (DeepStateMap) provides a visual representation of frontline developments by converting territorial change into geospatial figures. The module processes daily occupied-area polygons from DeepStateMap, reprojects them into a metric coordinate system, and calculates differences between consecutive days over the past week. Gains and losses are expressed as new geometries, each with associated area in square kilometers. The largest three gains and losses for each day are extracted and visualized on static maps, using red polygons for Russian advances and blue polygons for Ukrainian recaptures, overlaid on an OpenStreetMap basemap for geographic context. Each map includes an inset annotation of the areal extent of the change, while a centered title reports daily totals of territorial gains and losses. These daily map panels are structured as a chronological sequence of figures with headings summarizing the balance of territorial shifts for each day. The report provides both a high-level overview of weekly dynamics and a fine-grained spatial account of where control has changed. By integrating geospatial analysis with cartographic outputs, the module transforms numerical calculations of gains and losses into immediately interpretable visual evidence, ensuring that territorial trends are communicated in an accessible form. Limitations include the resolution of input polygons, which may generalize frontline positions, and the focus on the top three changes per side, which may omit smaller but operationally relevant shifts.

Module 7: Graphs and Pie Charts (DeepStateMap) complements the cartographic outputs with a quantitative visualization of territorial changes over the past week. The module calculates daily gains and losses by comparing consecutive occupied-area polygons, intersecting the resulting geometries with administrative oblast boundaries to attribute changes to specific regions. The outputs are aggregated into a structured dataset capturing Russian and Ukrainian area changes in square kilometers, disaggregated by date and oblast. This information is then transformed into a stacked bar chart showing the scale and composition of daily advances and retreats, with separate columns for Russian occupation and Ukrainian liberation. Distinctive colors assigned to each oblast allow for quick identification of where changes occurred, while patterned hatching differentiates Ukrainian gains from Russian gains. Complementing the time-series perspective, proportional pie charts summarize the distribution of total gains by oblast for each side, scaled in size according to overall magnitude. This dual visualization highlights both temporal dynamics and spatial concentration of activity, making it clear which regions contributed most to shifts in control. The module also generates a statistical summary of the week's developments, including total gains for each side, the three most affected oblasts, and the dates on which peak advances occurred. Limitations stem from the resolution of frontline data and the aggregation of small-scale fluctuations, which may mask micro-level tactical shifts while still capturing operational-level trends.

Module 8: Weekly Territorial Map (DeepStateMap) builds a weekly territorial overview by comparing occupied frontline geometries from the DeepStateMap dataset. The method first identifies the seven most recent days of data and establishes a baseline by selecting the last available date before this weekly window. For longer-term context, a secondary baseline is also taken from one year earlier. By applying geometric operations, the method calculates both newly occupied territories (Russian gains) and liberated areas (Ukrainian gains) over the past seven days. These areas are measured in square kilometers and ranked, allowing the identification of the three largest changes on each side. The visualization combines two layers of analysis: side panels zoom into the three most significant gains for Russia and Ukraine, while the central map shows the full Ukrainian theater. This central view highlights one-year-old occupation zones in gray, overlays recent gains in red and blue, and uses directional arrows to link the side maps with their precise locations. To ensure interpretability, all geometries are reprojected into equal-area coordinates for

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accurate measurement, and then into web-mercator projection for basemap rendering. The resulting figure situates short-term developments within the broader strategic picture, showing both the absolute scale of weekly gains and their spatial distribution across the front. A concise summary quantifies total Russian and Ukrainian gains, while the side panels and arrows guide the reader to the most dynamic hotspots of the week.

Module 9: TOP3 Territorial Maps (DeepStateMap) expands the territorial change analysis by focusing on the three largest weekly gains for each side and enriching them with predictive and contextual layers. After isolating the top three polygons of change, the method generates a three-by-three grid of maps: the first column displays current occupation zones overlaid on OpenStreetMap basemaps, the second column projects possible near-future contested areas by buffering the current gains and intersecting them with opposing control, and the third column adds a terrain perspective by draping these areas over a reprojected NASA digital elevation model. To enhance interpretability, each map includes dynamic scale bars, highlighted settlement markers, and frames that visually emphasize the change area. A narrative generator complements the maps by describing the absolute and relative scale of each gain, its share of the weekly total, and the settlements currently or potentially affected if the advance continues. This produces not only cartographic but also textual intelligence, making the output suitable for briefing contexts where both visual and written situational awareness are required.